

AI/ML Semantic Search Reference

Machine Learning

Machine learning enables computer systems to learn patterns from data and use those patterns to make predictions or decisions.

Supervised Learning

Supervised learning uses labeled training data. Classification predicts discrete categories, while regression predicts continuous numerical values.

Overfitting

Overfitting happens when a model learns the training data too closely and does not generalize well to unseen data.

Preventing Overfitting

Overfitting can be reduced using cross-validation, regularization, reducing unnecessary features, limiting model complexity, and collecting more representative training data.

Model Evaluation

Common evaluation measures include accuracy, precision, recall, and F1-score. A separate test set can be used to evaluate final performance on unseen data.

Feature Scaling

Feature scaling transforms numerical features to comparable ranges. It can be important for algorithms that depend on distances or feature magnitudes.

Decision Trees and Random Forests

Decision trees make predictions using feature-based splits. Random forests combine multiple decision trees to improve generalization.

Support Vector Machines

Support Vector Machines identify decision boundaries that separate classes. They can be used for classification tasks.

Neural Networks

Neural networks use layers of interconnected computational units to learn patterns from data.